

Plants in limbo — the theory of detrital photosynthesis

Luka Seamus Wright^{1,2*}

¹Oceans Institute, University of Western Australia, Perth, Australia

²School of Biological Sciences, University of Western Australia, Perth, Australia

*luka.wright@research.uwa.edu.au, luka@wright.it, orcid.org/0000-0002-1273-6256

Seagrass methods

Sample collection

Halophila ovalis ramets with several leaf pairs and intact roots were collected from Pelican Point in the brackish Swan Estuary (31.98715°S, 115.82101°E) alongside sediment from adjacent Matilda Bay (31.97572°S, 115.82324°E) on 15th September 2022. I filled the bottom of a 220-L glass flow-through seawater tank with the sediment and planted the ramets to acclimatise them to ambient seawater temperature and salinity and a 12-12-h light-dark cycle ($11 \pm 1.6 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$, mean $\pm$ standard deviation, cf. Masini & Manning 1997) for about two hours. *Amphibolis antarctica* stems with several leaf clusters were collected from fully marine Marmion Lagoon (31.85303°S, 115.75092°E) on 13th October 2022 and immediately used in the experiment. Sediment and seagrasses were collected from less than one metre depth.

Experiment

The *ex situ* decomposition experiment was commenced on the respective day of collection by excising individual leaves (*H. ovalis*) or leaf clusters (*A. antarctica*), hereinafter leaves, and immediately measuring photosynthesis (see next paragraph) of six (*H. ovalis*) or nine (*A. antarctica*) samples. Excision prematurely induces senescence and was intended to mimic detachment of non-senescent leaves, which may be caused by hydrodynamics or herbivory, rather than abscission. The remaining leaves were enclosed in mesh (3-mm $\varnothing$). Mesh bags were weighed down with inert aquarium pebbles and placed on the sediment in the same 220-L tank under identical photon fluence rate of $11 \pm 1.6 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$, which is sufficient for detrital photosynthesis to play a role (Wright et al. 2024) and is representative of the light environment experienced by detritus in the understory of kelp forests and seagrass meadows or on deeper sediment flats. To measure photosynthesis at each of five (*A. antarctica*) to six (*H. ovalis*) timepoints over 34 days following excision, six destructive samples were taken from the mesh bags which were kept submerged for the entire duration of the experiment.

Photosynthesis measurement

Light-saturated net photosynthesis (P_{max}) of leaves was measured using closed oxygen (O_2) incubations (cf. Wright et al. 2024). *A. antarctica* leaves weigh roughly ten times as much as *H. ovalis* leaves (given my data the mass ratio seems to be closer to 14:1). Since I wanted to measure both seagrasses at a comparable leaf mass per volume for the same duration, I incubated ten *H. ovalis* leaves or single *A. antarctica* leaves alongside seawater blanks in sealed glass jars filled with water collected from the flow-through system. Each jar was equipped with a magnetic stir bar and a self-adhesive planar O_2 sensor spot (SP-PS3-SA-NAU-D5-YOP, PreSens Precision Sensing GmbH, Regensburg, Germany) and placed on a magnetic stirrer under a saturating (Masini & Manning 1997, Jamaludin et al. 2006, Said et al. 2021) irradiance of $420 \pm 19 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$ (Zeus, Ledzeal, Shenzhen Topline Lighting Technology Co. Ltd., Shenzhen, China) in a 20°C room. Care was taken to exclude air from jars. Dissolved O_2 (μM) was measured fibre-optically through the glass every 10 s over at least 35 min with a four-channel O_2 meter (OXY-4 SMA G2, PreSens Precision Sensing GmbH, Regensburg, Germany). The O_2 meter was calibrated using anoxic (1% w/v Na_2SO_3) and air-saturated (bubbled with air) ultrapure water. All measurements were corrected for incubation temperature ($18 \pm 0.73^\circ\text{C}$), pressure ($1020 \pm 6 \text{ hPa}$) and salinity ($35 \pm 0.48 \text{ ‰}$) using a single temperature dipping probe connected to the first channel of the O_2

meter's built-in temperature sensor and placed in a fifth jar filled with seawater, the O₂ meter's built-in pressure sensor and a handheld refractometer respectively. I blotted and weighed (0.01-g accuracy, 440-33N, Kern & Sohn GmbH, Balingen, Germany) leaves and gravimetrically determined jar volume with ultrapure water to standardise $P_{\max}$ as shown in Equation S1.

Data analysis

In the first instance, simple linear models with centred incubation time ($t - \bar{t}$, min) as the predictor and dissolved O₂ (μM) as the response variable were fit to measurements from each sample and blank incubation. This yielded posterior probability distributions for slopes (β , $\mu\text{M min}^{-1}$) which were converted to mass-based $P_{\max}$ ($\mu\text{mol g}^{-1} \text{h}^{-1}$) as

$$P_{\max} = \frac{(\beta_s - \beta_b) \times V \times \Delta t}{m} \quad (\text{S1})$$

where subscript s and b denote sample and blank incubations from the same measurement group, V is the incubation volume (175 ± 0.17 mL) in litres, m is the sample blotted mass (0.52 ± 0.23 g) in grams and Δt is the desired period (1 h) in minutes. I also calculated leaf-based $P_{\max}$. For *A. antarctica* this was simply $(\beta_s - \beta_b) \times V \times \Delta t$. For *H. ovalis* the rate was additionally divided by 10 because there were 10 leaves in each incubation chamber.

Varying conditions during the 35-min incubations described above could affect $P_{\max}$ and confound the experimental effect. Possible confounding variables that I measured are initial incubation O₂ (μM), mean incubation temperature (T , °C), mean incubation pressure (P , hPA), incubation salinity (S , ‰) and sample mass (m , g). Although the oxygen measurements themselves were corrected for temperature, pressure and salinity, photosynthesis could also be influenced by these variables. Similarly, even though $P_{\max}$ was standardised by mass (Equation S1), the amount of leaf relative to water in the incubation chamber could additionally affect photosynthesis by changing the availability of carbon and nutrients. I explored these unwanted effects using a multiple linear regression of $P_{\max}$ against standardised (minus mean, divided by standard deviation) confounders:

$$\begin{aligned} P_{\max} &\sim \text{Normal}(\mu_{P_{\max}}, \sigma_{P_{\max}}) \\ \mu_{P_{\max}} &= \alpha + \beta_{O_2} \times O_2 + \beta_T \times T + \beta_P \times P + \beta_S \times S + \beta_m \times m \\ \alpha &\sim \text{Normal}(21, 10) \\ \beta_{O_2}, \beta_T, \beta_P, \beta_S, \beta_m &\sim \text{Normal}(0, 1) \\ \sigma_{P_{\max}} &\sim \text{Exponential}(1) \end{aligned} \quad (\text{S2})$$

The likelihood for $P_{\max}$ is normal because predictions of net photosynthesis can take negative values. Each parameter was stratified by species. Even though $P_{\max}$ was only weakly associated with confounders (Figure S1), I decided to account for them in the main model. To predict $P_{\max}$ with detrital age (t , days), I used the logistic model of detrital photosynthesis (Equation 1) as

$$\begin{aligned} P_{\max} &\sim \text{Normal}(\mu_{P_{\max}}, \sigma_{P_{\max}}) \\ \mu_{P_{\max}} &= \frac{\alpha + \tau}{1 + e^{\frac{5(t-\mu)}{\mu}}} - \tau \\ \ln \alpha &= \overline{\ln \alpha} + \beta_{\alpha O_2} \times O_2 + \beta_{\alpha T} \times T + \beta_{\alpha P} \times P + \beta_{\alpha S} \times S + \beta_{\alpha m} \times m \\ \ln \mu &= \overline{\ln \mu} + \beta_{\mu O_2} \times O_2 + \beta_{\mu T} \times T + \beta_{\mu P} \times P + \beta_{\mu S} \times S + \beta_{\mu m} \times m \\ \ln \tau &= \overline{\ln \tau} + \beta_{\tau O_2} \times O_2 + \beta_{\tau T} \times T + \beta_{\tau P} \times P + \beta_{\tau S} \times S + \beta_{\tau m} \times m \\ \overline{\ln \alpha} &\sim \text{Normal}(\ln 21, 0.4) \\ \overline{\ln \mu} &\sim \text{Normal}(\ln 14, 0.6) \\ \overline{\ln \tau} &\sim \text{Normal}(\ln 2, 0.4) \\ \beta_{\alpha O_2}, \beta_{\alpha T}, \beta_{\alpha P}, \beta_{\alpha S}, \beta_{\alpha m}, \beta_{\mu O_2}, \beta_{\mu T}, \beta_{\mu P}, \beta_{\mu S}, \beta_{\mu m}, \beta_{\tau O_2}, \beta_{\tau T}, \beta_{\tau P}, \beta_{\tau S}, \beta_{\tau m} &\sim \text{Normal}(0, 1) \\ \sigma_{P_{\max}} &\sim \text{Exponential}(1) \end{aligned} \quad (\text{S3})$$

where α is baseline photosynthesis ($\mu\text{mol g}^{-1} \text{h}^{-1}$), μ is the photosynthetic half-life (days) and τ is respiration of dead leaves ($\mu\text{mol g}^{-1} \text{h}^{-1}$). These parameters are strictly positive, so their logarithm is described by a multiple linear regression against the confounders (cf. Equation S2). Some species-specific influence of confounders on α and μ is evident (Figure S1), so their inclusion in this model is important for accurate prediction. I also ran this model for leaf-based P_{max} , adjusting the priors for α and τ accordingly.

Prior probability distributions for parameters were chosen based on published estimates. For the initial linear regressions of O_2 against incubation time, I centred the intercept prior on regional average seawater O_2 and the slope prior on zero. For α (Equation S2, S3), I chose the prior mean based on various published P_{max} estimates for *H. ovalis* ($40 \pm 57 \mu\text{mol g}^{-1} \text{h}^{-1}$, $1.8 \pm 2.5 \mu\text{mol leaf}^{-1} \text{h}^{-1}$, $n = 32$, Björk et al. 1997, Jamaludin et al. 2006, Borum et al. 2016, Lamit & Tanaka 2021, Said et al. 2021, 2025) and *A. antarctica* ($14 \pm 9 \mu\text{mol g}^{-1} \text{h}^{-1}$, $8.8 \pm 5.5 \mu\text{mol leaf}^{-1} \text{h}^{-1}$, $n = 8$, Masini & Manning 1997, Borum et al. 2016, Said et al. 2025). Units were converted where necessary by multiplying by mean dry-fresh mass ratios for *H. ovalis* ($0.21 \text{ g dry mass g}^{-1} \text{ fresh mass}$) and *A. antarctica* ($0.29 \text{ g dry mass g}^{-1} \text{ fresh mass}$) (de los Santos et al. 2012, Borum et al. 2016) and my own leaf fresh masses for *H. ovalis* ($0.04 \pm 0.01 \text{ g leaf}^{-1}$) and *A. antarctica* ($0.61 \pm 0.27 \text{ g leaf}^{-1}$). For μ (Equation S3), I settled on a median of 14 days, close to the minimum of 15 days for *Ecklonia radiata* (Wright et al. 2024) because I expected photosynthetic decay to be faster than for seaweeds. For τ (Equation S3), I centred the prior around published estimates of O_2 consumption by leaf litter of *Zostera marina* ($17 \pm 5.6 \mu\text{mol g}^{-1} \text{h}^{-1}$, $n = 16$, Blum & Mills 1991) and *Posidonia oceanica* ($1.3 \pm 1.3 \mu\text{mol g}^{-1} \text{h}^{-1}$, $n = 46$, Mateo & Romero 1996). Units were again converted by multiplying by mean dry-fresh mass ratios for *Z. marina* ($0.23 \text{ g dry mass g}^{-1} \text{ fresh mass}$, Evans et al. 1986) and *P. oceanica* ($0.24 \text{ g dry mass g}^{-1} \text{ fresh mass}$, Apostolaki et al. 2024). I chose a reasonable standard deviation for each prior based on variation in the published estimates and prior simulation.

The treatment of measurement error deserves brief mention. β and V (Equation S1) and hence P_{max} (Equation S1, S2, S3) as well as O_2 and T (Equation S2, S3) are measured with error, which I propagated into each downstream models as the standard deviation of upstream posteriors. This was achieved by parameterising latent variables for P_{max} , O_2 and T . In all cases I used the error structure $\text{mean} \sim \text{Normal}(\text{latent}, \text{standard deviation})$ where each observed mean is described by a normal distribution centred on the unobserved true value with observed error (McElreath 2020). For the response variable P_{max} , the latent variable is described by the normal likelihood (Equation S2, S3). For the standardised predictor variables O_2 and T , the latent variable is described by a standard normal prior: $\text{latent} \sim \text{Normal}(0, 1)$. Wherever a variable is measured with error, I visualised each observation as a distribution (violin or ellipse) rather than a point.

Data analysis and visualisation were performed in R v4.5.1 (R Core Team 2026) within the integrated development environment RStudio v2026.01.0+392 (RStudio Team 2026). General workflow was facilitated by the tidyverse package family v2.0.0 (Wickham et al. 2019) and here v1.0.2 (Müller 2025). Hamiltonian Monte Carlo models were written in Stan (Carpenter et al. 2017) and run with CmdStan v2.17.1 (Lee et al. 2017) via the R interface cmdstanr v0.9.0 (Gabry et al. 2025). All models were run with 8 Markov chains spread across all cores with 10^4 warmup and sampling iterations each. Convergence and smooth sampling were optimised by assessing $\hat{R}$ scores and visually scrutinising trace rank and pair plots with bayesplot v1.14.0 (Gabry et al. 2019). All reported results are posteriors and derived central tendencies and intervals calculated with tidybayes v3.0.7 (Kay 2024). Figure and table creation was facilitated by ggrridges v0.5.7 (Wilke 2021), ggh4x v0.3.1 (van den Brand 2025), geomtextpath v0.2.0 (Cameron & van den Brand 2022), patchwork v1.3.2 (Pedersen 2025) and officer v0.7.0 (Gohel et al. 2025). For detailed insight into data analysis and visualisation, all data, R and Stan code and diagnostic plots

can be consulted at github.com/lukaseamus/plant-limbo. Vector graphics were polished in Affinity Designer v1.10.8 (Serif Ltd., Nottingham, UK).

References

- Apostolaki ET, Lavery PS, Litsi-Mizan V, Serrano E, Inostroza K, Gerakaris V, Dailianis T, Glampedakis J, Holitzki T, Johnson E (2024) Patterns of carbon and nitrogen accumulation in seagrass (*Posidonia oceanica*) meadows of the Eastern Mediterranean Sea. *Journal of Geophysical Research: Biogeosciences* 129, e2024JG008163. <https://doi.org/10.1029/2024jg008163>.
- Björk M, Weil A, Semesi S, Beer S (1997) Photosynthetic utilisation of inorganic carbon by seagrasses from Zanzibar, East Africa. *Marine Biology* 129, 363–366. <https://doi.org/10.1007/s002270050176>.
- Blum LK, Mills AL (1991) Microbial growth and activity during the initial stages of seagrass decomposition. *Marine Ecology Progress Series* 70, 73–82. <https://doi.org/10.3354/meps070073>.
- Borum J, Pedersen O, Kotula L, Fraser MW, Statton J, Colmer TD, Kendrick GA (2016) Photosynthetic response to globally increasing CO₂ of co-occurring temperate seagrass species. *Plant, Cell & Environment* 39, 1240–1250. <https://doi.org/10.1111/pce.12658>.
- van den Brand T (2025) ggh4x: hacks for 'ggplot2'. <https://github.com/teunbrand/ggh4x>.
- Cameron A, van den Brand T (2022) geomtextpath: use curved text in ggplot2. R package v0.1.0. <https://allanccameron.github.io/geomtextpath>.
- Carpenter B, Gelman A, Hoffman MD, Lee D, Goodrich B, Betancourt M, Brubaker M, Guo J, Li P, Riddell A (2017) Stan: a probabilistic programming language. *Journal of Statistical Software* 76, 1–32. <https://doi.org/10.18637/jss.v076.i01>.
- Evans AS, Webb KL, Penhale PA (1986) Photosynthetic temperature acclimation in two coexisting seagrasses, *Zostera marina* L. and *Ruppia maritima* L. *Aquatic Botany* 24, 185–197. [https://doi.org/10.1016/0304-3770\(86\)90095-1](https://doi.org/10.1016/0304-3770(86)90095-1).
- Gabry J, Simpson D, Vehtari A, Betancourt M, Gelman A (2019) Visualization in Bayesian workflow. *Journal of the Royal Statistical Society Series A: Statistics in Society* 182, 389–402. <https://doi.org/10.1111/rssa.12378>.
- Gabry J, Češnovar R, Johnson A, Bröder S (2025) cmdstanr: R Interface to 'CmdStan'. <https://mc-stan.org/cmdstanr>.
- Gohel D, Moog S, Heckmann M (2025) officer: manipulation of Microsoft Word and PowerPoint documents. <https://ardata-fr.github.io/officerverse>.
- Jamaludin MR, Bujang JS, Kusnan M, Omar HO (2006) Photosynthetic light responses of wild and cultured *Halophila ovalis*. *Malaysian Journal of Science* 25, 67–79.
- Kay M (2024) tidybayes: tidy data and geoms for Bayesian models. <https://doi.org/10.5281/zenodo.13770114>.
- Lamit N, Tanaka Y (2021) Effects of river water inflow on the growth, photosynthesis, and respiration of the tropical seagrass *Halophila ovalis*. *Botanica Marina* 64, 93–100. <https://doi.org/10.1515/bot-2020-0079>.
- Lee D, Stan buildbot, seantalts, Carpenter B, Morris M, Kucukelbir A, Betancourt M, Tran D, Brubaker M, Bales B et al. (2017) stan-dev/cmdstan: v2.17.1. <https://doi.org/10.5281/zenodo.598253>.
- Masini RJ, Manning CR (1997) The photosynthetic responses to irradiance and temperature of four meadow-forming seagrasses. *Aquatic Botany* 58, 21–36. [https://doi.org/10.1016/s0304-3770\(97\)00008-9](https://doi.org/10.1016/s0304-3770(97)00008-9).
- Mateo MA, Romero J (1996) Evaluating seagrass leaf litter decomposition: an experimental comparison between litter-bag and oxygen-uptake methods. *Journal of Experimental Marine Biology and Ecology* 202, 97–106. [https://doi.org/10.1016/0022-0981\(96\)00019-6](https://doi.org/10.1016/0022-0981(96)00019-6).
- McElreath R (2020) Statistical rethinking²: a Bayesian course with examples in R and Stan, Second Edition. Chapman & Hall/CRC. <https://doi.org/10.1201/9780429029608>.

- Müller K (2025) here: a simpler way to find your files. <https://here.r-lib.org>.
- Pedersen TL (2025) patchwork: the composer of plots. <https://patchwork.data-imaginist.com>.
- R Core Team (2026) R: a language and environment for statistical computing. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- RStudio Team (2026) RStudio: integrated development environment for R. RStudio, Inc. <http://www.rstudio.com/>.
- Said NE, McMahon K, Lavery PS (2021) Accounting for the influence of temperature and location when predicting seagrass (*Halophila ovalis*) photosynthetic performance. *Estuarine, Coastal and Shelf Science* 257, 107414. <https://doi.org/10.1016/j.ecss.2021.107414>.
- Said N, Webster C, Strydom S, Dunham N, Adams MP, McMahon K (2025) Seagrasses are most vulnerable to marine heatwaves in tropical zones: local-scale and broad climatic zone variation in thermal tolerances. *New Phytologist*. <https://doi.org/10.1111/nph.70742>.
- de los Santos CB, Brun FG, Onoda Y, Cambridge ML, Bouma TJ, Vergara JJ, Pérez-Lloréns JL (2012) Leaf-fracture properties correlated with nutritional traits in nine Australian seagrass species: implications for susceptibility to herbivory. *Marine Ecology Progress Series* 458, 89–102. <https://doi.org/10.3354/meps09757>.
- Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R, Grolemond G, Hayes A, Henry L, Hester J (2019) Welcome to the Tidyverse. *Journal of Open Source Software* 4, 1686. <https://doi.org/10.21105/joss.01686>.
- Wilke CO (2021) gggridges: ridgeline plots in ggplot2. <https://CRAN.R-project.org/package=gggridges>.
- Wright LS, Simpkins T, Filbee-Dexter K, Wernberg T (2024) Temperature sensitivity of detrital photosynthesis. *Annals of Botany* 133, 17–28. <https://doi.org/10.1093/aob/mcad167>.

Figures

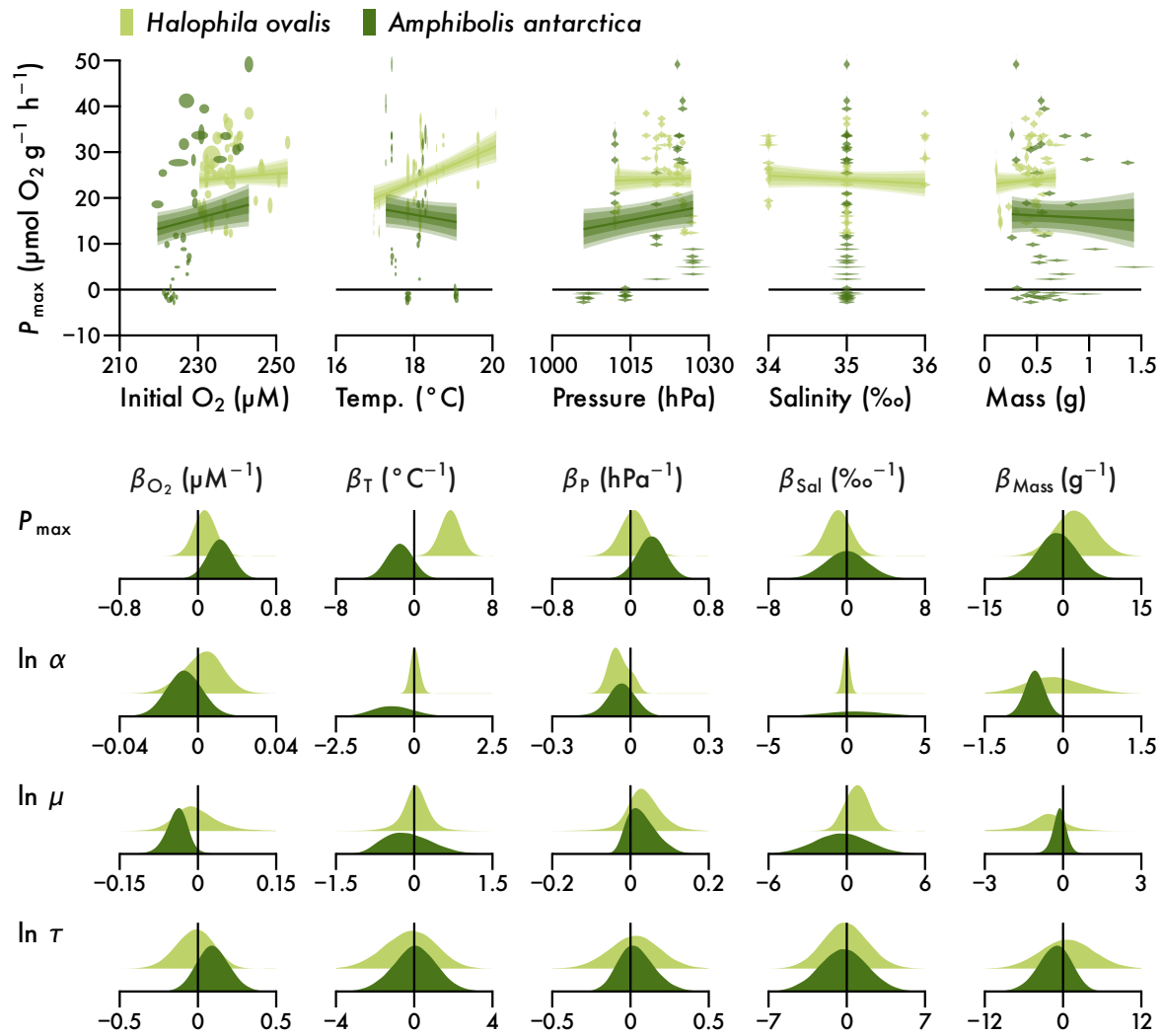


Figure S1. Effect of confounding variables on light-saturated net photosynthesis ($P_{\max}$) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves. $P_{\max}$ is standardised by leaf fresh mass. Ellipses and violins are kernel density estimates of observations with measurement error ($n = 42$ for *H. ovalis*, $n = 39$ for *A. antarctica*). Lines and intervals are medians and the central 50%, 80% and 90% of posterior probability for predicted means. Distributions are kernel density estimates of slope posteriors (Equation S2, S3).

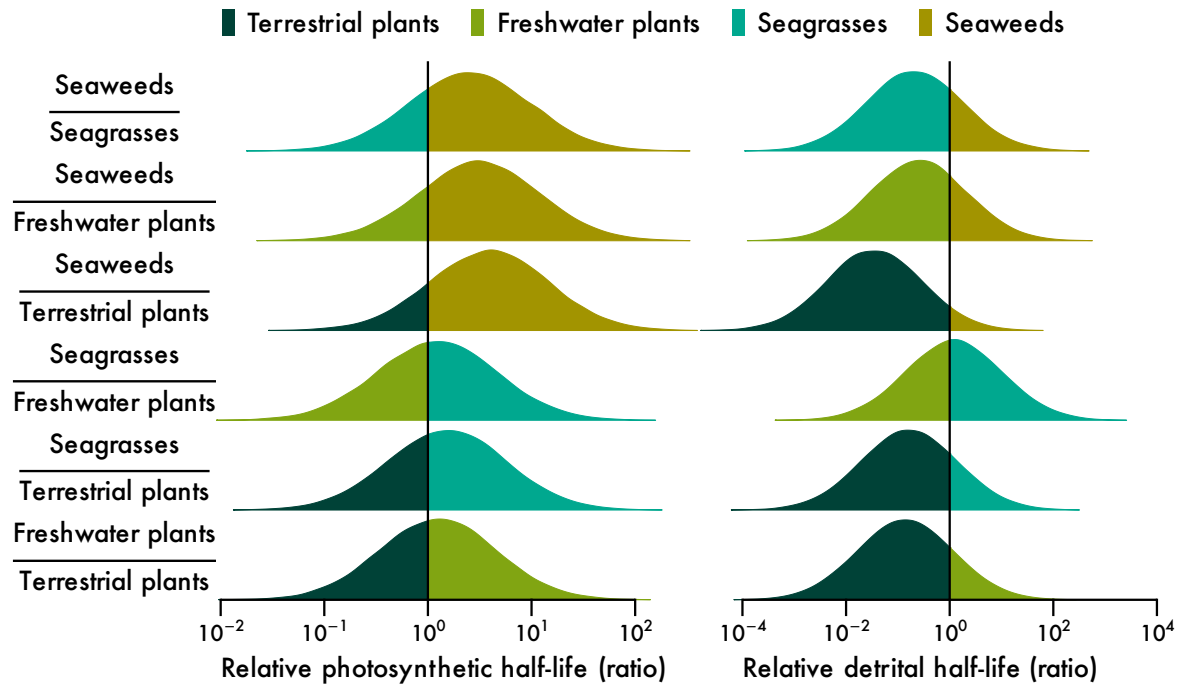


Figure S2. Photosynthetic and detrital half-life contrasts between plant groups. The group with more probability (coloured area) either side of the identity line at ratio = 1 has a greater half-life.

Tables

Table S1. Species included in the meta-analysis. Phyla are given in bold and orders are underlined. Seaweeds are comprised of Chlorophyta and Heterokontophyta. Seagrasses are marked with an asterisk. All remaining Alismatales are freshwater plants and the rest of Streptophyta is terrestrial. The longest experimental durations and largest numbers of studies, experiments and observations in each non-taxonomic group are highlighted in bold.

	Species	Days	Studies	Exper.	Observ.
Chlorophyta					
<u>Ulvales</u>					
Ulvaceae	<i>Ulva clathrata</i>	30	1	1	6
	<i>Ulva intestinalis</i>	30	1	1	6
	<i>Ulva reticulata</i>	30	1	1	6
Heterokontophyta					
<u>Desmarestiales</u>					
Desmarestiaceae	<i>Desmarestia anceps</i>	313	1	1	65
<u>Fucales</u>					
Durvillaeaceae	<i>Durvillaea antarctica</i>	14	2	16	448
Sargassaceae	<i>Sargassum fallax</i>	14	1	9	84
	<i>Sargassum spinuligerum</i>	14	1	13	159
Seirococcaceae	<i>Scytothalia dorycarpa</i>	14	1	6	48
<u>Laminariales</u>					
Laminariaceae	<i>Laminaria digitata</i>	46	4	6	191
	<i>Laminaria hyperborea</i>	168	7	56	559
	<i>Laminaria ochroleuca</i>	56	4	51	416
	<i>Macrocystis pyrifera</i>	55	4	23	375
	<i>Saccharina latissima</i>	39	2	2	55
Lessoniaceae	<i>Ecklonia radiata</i>	119	2	10	117
<u>Tilopteridales</u>					
Phyllariaceae	<i>Saccorhiza polyschides</i>	39	1	1	38
Streptophyta					
<u>Alismatales</u>					
Cymodoceaceae	<i>Amphibolis antarctica</i> *	34	1	1	39
Hydrocharitaceae	<i>Elodea densa</i>	16	2	3	43
	<i>Halophila ovalis</i> *	34	1	1	42
	<i>Hydrilla verticillata</i>	12	7	16	506
	<i>Thalassia testudinum</i> *	52	1	1	14
	<i>Vallisneria americana</i>	104	1	1	5
	<i>Vallisneria spiralis</i>	9	6	6	59
Potamogetonaceae	<i>Potamogeton</i> sp.	104	1	1	5
	<i>Stuckenia pectinata</i>	9	5	5	55
Zosteraceae	<i>Zostera marina</i> *	85	1	1	7
	<i>Zostera muelleri</i> *	16	1	4	58
<u>Apiales</u>					
Apiaceae	<i>Coriandrum sativum</i>	9	1	1	24
	<i>Petroselinum crispum</i>	7	1	1	7
<u>Asterales</u>					
Asteraceae	<i>Chrysanthemum</i> × <i>morifolium</i>	12	1	2	30
	<i>Helianthus annuus</i>	0.024	1	1	36
	<i>Helianthus tuberosus</i>	0.026	1	1	36
	<i>Inula racemosa</i>	0.025	1	1	36
	<i>Smallanthus connatus</i>	0.026	1	1	36

<u>Brassicales</u>					
Brassicaceae	<i>Arabidopsis thaliana</i>	6	2	3	60
	<i>Brassica napus</i>	1.7	1	25	125
	<i>Brassica oleracea</i>	10	1	3	22
	<i>Nasturtium officinale</i>	5	2	3	39
	<i>Raphanus raphanistrum</i>	6	1	4	20
	<i>Sinapis alba</i>	6	1	4	20
Tropaeolaceae	<i>Tropaeolum majus</i>	8	2	3	27
<u>Caryophyllales</u>					
Amaranthaceae	<i>Amaranthus cruentus</i>	0.026	1	1	36
	<i>Celosia argentea</i>	0.024	1	1	36
	<i>Chenopodium berlandieri</i>	0.27	1	2	26
	<i>Gomphrena serrata</i>	0.024	1	1	36
	<i>Spinacia oleracea</i>	10	3	4	85
Polygonaceae	<i>Koenigia weyrichii</i>	0.028	1	1	36
<u>Cucurbitales</u>					
Cucurbitaceae	<i>Cucumis sativus</i>	13	2	3	13
	<i>Sicyos edulis</i>	10	1	2	48
<u>Ericales</u>					
Theaceae	<i>Camellia sinensis</i>	4	1	2	11
<u>Fabales</u>					
Fabaceae	<i>Cercis canadensis</i>	0.0052	1	4	576
	<i>Lathyrus oleraceus</i>	17	2	2	42
	<i>Phaseolus vulgaris</i>	2.1	1	1	10
	<i>Trifolium subterraneum</i>	7	2	5	25
	<i>Vigna unguiculata</i>	4	1	10	170
<u>Fagales</u>					
Betulaceae	<i>Betula pendula</i>	0.029	1	1	36
Fagaceae	<i>Quercus alba</i>	0.01	1	1	24
	<i>Quercus muehlenbergii</i>	0.0052	1	1	192
	<i>Quercus robur</i>	0.024	2	2	228
	<i>Quercus rubra</i>	0.01	1	1	24
Juglandaceae	<i>Juglans nigra</i>	0.01	1	1	24
<u>Lamiales</u>					
Lamiaceae	<i>Salvia officinalis</i>	11	1	1	8
<u>Laurales</u>					
Lauraceae	<i>Cinnamomum tamala</i>	21	1	1	36
<u>Pinales</u>					
Pinaceae	<i>Larix sibirica</i>	0.025	1	1	36
	<i>Pinus sibirica</i>	0.023	1	1	33
	<i>Pinus sylvestris</i>	0.022	1	1	33
<u>Poales</u>					
Cyperaceae	<i>Bolboschoenus fluviatilis</i>	104	1	1	5
Poaceae	<i>Avena sativa</i>	8	8	28	161
	<i>Hordeum vulgare</i>	11	13	25	466
	<i>Lolium multiflorum</i>	5	1	3	44
	<i>Lolium pratense</i>	7	5	8	112
	<i>Oryza sativa</i>	27	11	48	327
	<i>Panicum miliaceum</i>	5	2	2	69
	<i>Secale cereale</i>	5	2	9	102
	<i>Triticum aestivum</i>	10	13	41	646
	<i>Triticum turgidum</i>	6	1	1	7

	<i>Triticum vulgare</i>	6	1	2	48
	<i>Zea mays</i>	6	3	4	58
Typhaceae	<i>Typha angustifolia</i>	104	1	1	5
Polypodiales					
Nephrolepidaceae	<i>Nephrolepis exaltata</i>	10	1	4	24
<u>Rosales</u>					
Rosaceae	<i>Fragaria × ananassa</i>	0.0056	1	1	663
	<i>Malus domestica</i>	21	3	6	150
	<i>Malus hupehensis</i>	15	1	1	18
<u>Sapindales</u>					
Rutaceae	<i>Citrus × aurantium</i>	10	1	1	5
Sapindaceae	<i>Acer saccharinum</i>	0.25	1	4	36
	<i>Acer truncatum</i>	0.0052	1	1	192
<u>Solanales</u>					
Convolvulaceae	<i>Ipomoea batatas</i>	3	1	2	24
Solanaceae	<i>Nicotiana rustica</i>	10	1	2	180
	<i>Nicotiana tabacum</i>	9	1	2	24
	<i>Solanum lycopersicum</i>	5	1	6	192
	<i>Solanum melongena</i>	16	1	1	24
	<i>Solanum tuberosum</i>	0.026	1	1	36
<u>Vitales</u>					
Vitaceae	<i>Vitis vinifera</i>	0.0052	1	5	1280

Table S2. Predictions of photosynthetic half-life (days, μ in Equation 2) for new experiments on unobserved species in each plant group in this study as well as new plant groups. Estimates are stratified by observed differences in light condition and the type of response variable. All estimates are given as median (ln mean $\pm$ standard deviation).

Light	Response	Photosynthetic half-life
Terrestrial plants		
Light	Photosynthesis	13 (2.6 $\pm$ 0.94)
	Chlorophyll	18 (2.9 $\pm$ 0.93)
Dark	Photosynthesis	7.2 (2 $\pm$ 0.92)
	Chlorophyll	9.5 (2.3 $\pm$ 0.92)
Freshwater plants		
Light	Chlorophyll	19 (2.9 $\pm$ 0.98)
Dark	Photosynthesis	11 (2.4 $\pm$ 0.95)
	Chlorophyll	16 (2.8 $\pm$ 0.95)
Seagrasses		
Light	Photosynthesis	19 (2.9 $\pm$ 0.98)
	Chlorophyll	26 (3.3 $\pm$ 0.98)
Dark	Chlorophyll	16 (2.8 $\pm$ 1)
Seaweeds		
Light	Photosynthesis	51 (3.9 $\pm$ 0.94)
	Chlorophyll	81 (4.4 $\pm$ 0.96)
Dark	Photosynthesis	24 (3.2 $\pm$ 1)
All plants		
Light	Photosynthesis	17 (2.8 $\pm$ 1.1)
	Chlorophyll	24 (3.2 $\pm$ 1.1)
Dark	Photosynthesis	12 (2.4 $\pm$ 1.1)
	Chlorophyll	16 (2.8 $\pm$ 1.1)

Table S3. Predictions of photosynthetic half-life (days, μ in Equation 2) for new experiments on each species in this study (cf. Table S1 for complete taxonomy). Estimates are stratified by observed differences in light condition and the type of response variable. All estimates are given as median (ln mean $\pm$ standard deviation).

Species	Light	Response	Photosynthetic half-life
Seaweeds			
<i>Desmarestia anceps</i>	Light	Chlorophyll	154 (5 $\pm$ 0.9)
<i>Durvillaea antarctica</i>	Light	Photosynthesis	59 (4.1 $\pm$ 0.87)
		Chlorophyll	95 (4.6 $\pm$ 0.9)
<i>Ecklonia radiata</i>	Light	Photosynthesis	32 (3.5 $\pm$ 0.78)
	Dark	Photosynthesis	15 (2.7 $\pm$ 0.87)
<i>Laminaria digitata</i>	Light	Photosynthesis	68 (4.2 $\pm$ 0.86)
		Chlorophyll	109 (4.7 $\pm$ 0.88)
<i>Laminaria hyperborea</i>	Light	Photosynthesis	62 (4.1 $\pm$ 0.73)
		Chlorophyll	99 (4.6 $\pm$ 0.79)
<i>Laminaria ochroleuca</i>	Light	Photosynthesis	68 (4.2 $\pm$ 0.74)
		Chlorophyll	108 (4.7 $\pm$ 0.79)
<i>Macrocystis pyrifera</i>	Light	Photosynthesis	60 (4.1 $\pm$ 0.84)
		Chlorophyll	96 (4.6 $\pm$ 0.86)
<i>Saccharina latissima</i>	Light	Photosynthesis	60 (4.1 $\pm$ 0.9)
		Chlorophyll	96 (4.6 $\pm$ 0.91)
<i>Saccorhiza polyschides</i>	Light	Chlorophyll	87 (4.5 $\pm$ 0.92)
<i>Sargassum fallax</i>	Light	Photosynthesis	30 (3.4 $\pm$ 0.82)
<i>Sargassum spinuligerum</i>	Light	Photosynthesis	64 (4.2 $\pm$ 0.86)
		Chlorophyll	102 (4.6 $\pm$ 0.89)
<i>Scytothalia dorycarpa</i>	Light	Photosynthesis	45 (3.8 $\pm$ 0.95)
<i>Ulva clathrata</i>	Light	Chlorophyll	85 (4.4 $\pm$ 0.94)
<i>Ulva intestinalis</i>	Light	Chlorophyll	84 (4.4 $\pm$ 0.94)
<i>Ulva reticulata</i>	Light	Chlorophyll	85 (4.4 $\pm$ 0.94)
Seagrasses			
<i>Amphibolis antarctica</i>	Light	Photosynthesis	14 (2.6 $\pm$ 0.88)
<i>Halophila ovalis</i>	Light	Photosynthesis	27 (3.3 $\pm$ 0.91)
<i>Thalassia testudinum</i>	Light	Chlorophyll	26 (3.2 $\pm$ 0.87)
<i>Zostera marina</i>	Light	Chlorophyll	26 (3.3 $\pm$ 0.87)
<i>Zostera muelleri</i>	Light	Chlorophyll	25 (3.2 $\pm$ 0.86)
	Dark	Chlorophyll	16 (2.8 $\pm$ 0.86)
Freshwater plants			
<i>Elodea densa</i>	Light	Chlorophyll	18 (2.9 $\pm$ 0.89)
	Dark	Chlorophyll	15 (2.7 $\pm$ 0.86)
<i>Hydrilla verticillata</i>	Light	Chlorophyll	14 (2.6 $\pm$ 0.79)
	Dark	Photosynthesis	7.7 (2 $\pm$ 0.78)
	Dark	Chlorophyll	12 (2.5 $\pm$ 0.78)
<i>Potamogeton sp.</i>	Dark	Chlorophyll	25 (3.2 $\pm$ 0.88)
<i>Stuckenia pectinata</i>	Dark	Photosynthesis	7.1 (2 $\pm$ 0.8)
	Dark	Chlorophyll	11 (2.4 $\pm$ 0.82)
<i>Vallisneria americana</i>	Dark	Chlorophyll	25 (3.2 $\pm$ 0.88)
<i>Vallisneria spiralis</i>	Dark	Photosynthesis	7.8 (2.1 $\pm$ 0.79)
	Dark	Chlorophyll	12 (2.5 $\pm$ 0.81)
Terrestrial plants			
<i>Acer saccharinum</i>	Light	Photosynthesis	13 (2.6 $\pm$ 0.93)
<i>Acer truncatum</i>	Light	Photosynthesis	13 (2.6 $\pm$ 0.94)
<i>Amaranthus cruentus</i>	Light	Photosynthesis	13 (2.6 $\pm$ 0.93)

<i>Arabidopsis thaliana</i>	Dark	Photosynthesis	4.3 (1.5 ± 0.81)
	Dark	Chlorophyll	5.8 (1.8 ± 0.8)
<i>Avena sativa</i>	Light	Chlorophyll	11 (2.4 ± 0.78)
	Dark	Chlorophyll	5.8 (1.8 ± 0.76)
<i>Betula pendula</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Bolboschoenus fluviatilis</i>	Dark	Chlorophyll	20 (3 ± 0.9)
<i>Brassica napus</i>	Light	Chlorophyll	18 (2.9 ± 0.91)
<i>Brassica oleracea</i>	Dark	Chlorophyll	8.9 (2.2 ± 0.85)
<i>Camellia sinensis</i>	Light	Photosynthesis	14 (2.7 ± 0.91)
<i>Celosia argentea</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Cercis canadensis</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Chenopodium berlandieri</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Chrysanthemum × morifolium</i>	Dark	Chlorophyll	9.4 (2.2 ± 0.82)
<i>Cinnamomum tamala</i>	Dark	Chlorophyll	9.6 (2.3 ± 0.84)
<i>Citrus × aurantium</i>	Light	Chlorophyll	19 (2.9 ± 0.9)
<i>Coriandrum sativum</i>	Dark	Chlorophyll	9.6 (2.3 ± 0.87)
<i>Cucumis sativus</i>	Light	Chlorophyll	19 (3 ± 0.89)
	Dark	Chlorophyll	10 (2.3 ± 0.88)
<i>Fragaria × ananassa</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Gomphrena serrata</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Helianthus annuus</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Helianthus tuberosus</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Hordeum vulgare</i>	Light	Photosynthesis	9.2 (2.2 ± 0.77)
		Chlorophyll	12 (2.5 ± 0.76)
	Dark	Photosynthesis	5 (1.6 ± 0.75)
<i>Inula racemosa</i>		Chlorophyll	6.6 (1.9 ± 0.75)
	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Ipomoea batatas</i>	Dark	Photosynthesis	7.9 (2.1 ± 0.9)
		Chlorophyll	11 (2.4 ± 0.9)
<i>Juglans nigra</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Koenigia weyrichii</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Larix sibirica</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Lathyrus oleraceus</i>	Light	Photosynthesis	16 (2.8 ± 0.89)
		Chlorophyll	22 (3.1 ± 0.88)
<i>Lolium multiflorum</i>	Light	Photosynthesis	12 (2.5 ± 0.92)
		Chlorophyll	16 (2.8 ± 0.92)
<i>Lolium pratense</i>	Dark	Photosynthesis	4.1 (1.4 ± 0.77)
		Chlorophyll	5.5 (1.7 ± 0.77)
<i>Malus domestica</i>	Light	Photosynthesis	24 (3.2 ± 0.86)
		Chlorophyll	33 (3.5 ± 0.85)
	Dark	Photosynthesis	13 (2.6 ± 0.85)
<i>Malus hupehensis</i>		Chlorophyll	18 (2.9 ± 0.85)
	Dark	Chlorophyll	13 (2.6 ± 0.88)
<i>Nasturtium officinale</i>	Dark	Chlorophyll	4.3 (1.5 ± 0.81)
<i>Nephrolepis exaltata</i>	Light	Photosynthesis	18 (2.9 ± 0.87)
		Chlorophyll	24 (3.2 ± 0.86)
	Dark	Photosynthesis	9.8 (2.3 ± 0.86)
<i>Nicotiana rustica</i>		Chlorophyll	13 (2.6 ± 0.85)
	Dark	Photosynthesis	8.5 (2.1 ± 0.84)
<i>Nicotiana tabacum</i>		Chlorophyll	11 (2.4 ± 0.83)
	Light	Photosynthesis	16 (2.8 ± 0.89)
		Chlorophyll	21 (3.1 ± 0.88)

<i>Oryza sativa</i>	Light	Photosynthesis	10 (2.4 ± 0.77)
		Chlorophyll	14 (2.6 ± 0.76)
	Dark	Photosynthesis	5.7 (1.7 ± 0.76)
		Chlorophyll	7.5 (2 ± 0.75)
<i>Panicum miliaceum</i>	Dark	Chlorophyll	5.4 (1.7 ± 0.82)
<i>Petroselinum crispum</i>	Dark	Chlorophyll	10 (2.3 ± 0.89)
<i>Phaseolus vulgaris</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Pinus sibirica</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Pinus sylvestris</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Quercus alba</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Quercus muehlenbergii</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Quercus robur</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Quercus rubra</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Raphanus raphanistrum</i>	Light	Photosynthesis	14 (2.7 ± 0.89)
		Chlorophyll	19 (3 ± 0.89)
<i>Salvia officinalis</i>	Dark	Chlorophyll	12 (2.5 ± 0.89)
<i>Secale cereale</i>	Light	Photosynthesis	13 (2.6 ± 0.86)
		Chlorophyll	18 (2.9 ± 0.85)
	Dark	Photosynthesis	7.2 (2 ± 0.85)
		Chlorophyll	9.5 (2.3 ± 0.85)
<i>Sicyos edulis</i>	Dark	Photosynthesis	7.4 (2 ± 0.82)
		Chlorophyll	9.9 (2.3 ± 0.82)
<i>Sinapis alba</i>	Light	Photosynthesis	15 (2.7 ± 0.89)
		Chlorophyll	20 (3 ± 0.89)
<i>Smallanthus connatus</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Solanum lycopersicum</i>	Light	Photosynthesis	11 (2.4 ± 0.85)
<i>Solanum melongena</i>	Light	Photosynthesis	15 (2.7 ± 0.88)
<i>Solanum tuberosum</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Spinacia oleracea</i>	Dark	Photosynthesis	5.8 (1.8 ± 0.8)
		Chlorophyll	7.7 (2 ± 0.79)
<i>Trifolium subterraneum</i>	Light	Photosynthesis	12 (2.5 ± 0.88)
		Chlorophyll	16 (2.8 ± 0.88)
<i>Triticum aestivum</i>	Light	Photosynthesis	5.5 (1.7 ± 0.75)
		Chlorophyll	7.3 (2 ± 0.75)
	Dark	Photosynthesis	3 (1.1 ± 0.74)
		Chlorophyll	4 (1.4 ± 0.74)
<i>Triticum turgidum</i>	Light	Chlorophyll	18 (2.9 ± 0.92)
<i>Triticum vulgare</i>	Light	Photosynthesis	13 (2.6 ± 0.88)
	Dark	Photosynthesis	7.3 (2 ± 0.87)
<i>Tropaeolum majus</i>	Light	Chlorophyll	20 (3 ± 0.87)
	Dark	Chlorophyll	11 (2.4 ± 0.87)
<i>Typha angustifolia</i>	Dark	Chlorophyll	22 (3.1 ± 0.91)
<i>Vigna unguiculata</i>	Dark	Photosynthesis	5.3 (1.7 ± 0.78)
		Chlorophyll	7.2 (2 ± 0.78)
<i>Vitis vinifera</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Zea mays</i>	Light	Photosynthesis	14 (2.6 ± 0.91)
		Chlorophyll	19 (2.9 ± 0.9)
	Dark	Chlorophyll	10 (2.3 ± 0.89)

Meta-analysis references

Studies included in the meta-analysis in addition to the present study. Note that 130 references are listed because the data from three studies were not independent and thus merged.

- Albright LJ, Chocair J, Masuda K, Valdes M (1982) Degradation of the kelps *Macrocystis integrifolia* and *Nereocystis luetkeana* in British Columbia coastal waters. Srivastava LM (ed.) *Synthetic and degradative processes in marine macrophytes*, 215–234. <https://doi.org/10.1515/9783110837988-015>.
- Arguelles T, Guardiola JL (1977) Hormonal control of senescence in excised orange leaves. *Journal of Horticultural Science* 52, 199–204. <https://doi.org/10.1080/00221589.1977.11514748>.
- Arnold KE, Manley SL (1985) Carbon allocation in *Macrocystis pyrifera* (Phaeophyta): intrinsic variability in photosynthesis and respiration. *Journal of Phycology* 21, 154–167. <https://doi.org/10.1111/j.0022-3646.1985.00154.x>.
- Asim M, Guo M, Khan R, Sun Y, Du S, Liu W, Li Y, Wang X, Wang M, Shi Y et al. (2022) Investigation of sugar signaling behaviors involved in sucrose-induced senescence initiation and progression in *N. tabacum*. *Plant Physiology and Biochemistry* 184, 112–125. <https://doi.org/10.1016/j.plaphy.2022.05.024>.
- Barua DN (1960) Effect of age and carbon-dioxide concentration on assimilation by detached leaves of tea and sunflower. *The Journal of Agricultural Science* 55, 413–421. <https://doi.org/10.1017/s0021859600023327>.
- de Bettignies F, Dauby P, Thomas F, Gobet A, Delage L, Bohner O, Loisel S, Davoult D (2020) Degradation dynamics and processes associated with the accumulation of *Laminaria hyperborea* (Phaeophyceae) kelp fragments: an in situ experimental approach. *Journal of Phycology* 56, 1481–1492. <https://doi.org/10.1111/jpy.13041>.
- Bianchi TS, Findlay S (1991) Decomposition of Hudson estuary macrophytes: photosynthetic pigment transformations and decay constants. *Estuaries* 14, 65–73. <https://doi.org/10.2307/1351983>.
- Birecka H, Ireton KP, Bitonti AJ, McCann PP (1991) Endogenous polyamine levels and dark-induced leaf senescence. *Phytochemistry* 30, 105–108. [https://doi.org/10.1016/0031-9422\(91\)84107-4](https://doi.org/10.1016/0031-9422(91)84107-4).
- Biswal B, Choudhury NK, Prativa S, Biswal UC (1983) Senescence of detached fern leaves. *Plant and Cell Physiology* 24, 1203–1208. <https://doi.org/10.1093/oxfordjournals.pcp.a076634>.
- Biswal B, Joshi PN, Kulandaivelu G (1997) Changes in leaf protein and pigment contents and photosynthetic activities during senescence of detached maize leaves: influence of different ultraviolet radiations. *Photosynthetica* 34, 37–44. <https://doi.org/10.1023/a:1006855532424>.
- Biswal UC, Mohanty P (1976a) Dark stress induced senescence of detached barley leaves: II. Alteration in the absorption characteristic and photochemical activity of the chloroplasts isolated from senescing leaves. *Plant Science Letters* 7, 371–379. [https://doi.org/10.1016/0304-4211\(76\)90094-8](https://doi.org/10.1016/0304-4211(76)90094-8).
- Biswal UC, Mohanty P (1976b) Aging induced changes in photosynthetic electron transport of detached barley leaves. *Plant and Cell Physiology* 17, 323–331. <https://doi.org/10.1093/oxfordjournals.pcp.a075285>.
- Biswas AK, Choudhuri MA (1980) Mechanism of monocarpic senescence in rice. *Plant Physiology* 65, 340–345. <https://doi.org/10.1104/pp.65.2.340>.
- Biswas R, Sinhababu A, Banerjee A, Kar RK (2005) Qualitative and quantitative effect of light on leaf senescence in a submerged aquatic weed, *Hydrilla verticillata*. *Indian Journal of Plant Physiology* 10, 66.

- Bouma D (1975) Effects of some metabolic phosphorus compounds on rates of photosynthesis of detached phosphorus-deficient subterranean clover leaves. *Journal of Experimental Botany* 26, 52–59. <https://doi.org/10.1093/jxb/26.1.52>.
- Bouma D, Dowling EJ (1982) The relationship between the phosphorus status of subterranean clover plants and losses in chlorophyll from detached leaves: a basis for the visual diagnosis of phosphorus deficiency. *Annals of Botany* 49, 637–648. <https://doi.org/10.1093/oxfordjournals.aob.a086291>.
- Brady CJ, Heng FT (1975) Rate of protein synthesis in senescing, detached wheat leaves. *Functional Plant Biology* 2, 163–176. <https://doi.org/10.1071/pp9750163>.
- Brouwer PEM (1996) Decomposition in situ of the sublittoral Antarctic macroalga *Desmarestia anceps* Montagne. *Polar Biology* 16, 129–137. <https://doi.org/10.1007/bf02390433>.
- Causin HF, Marchetti CF, Pena LB, Gallego SM, Barneix AJ (2015) Down-regulation of catalase activity contributes to senescence induction in wheat leaves exposed to shading stress. *Biologia Plantarum* 59, 154–162. <https://doi.org/10.1007/s10535-014-0480-z>.
- Chalise R, Dahal A, Basnet S, Sharma S, Pant DR, Khanal R (2024) Effect of plasma-activated water on chlorophyll retention in detached Tejpat (*Cinnamomum tamala*) leaves. *Heliyon* 10, e24480. <https://doi.org/10.1016/j.heliyon.2024.e24480>.
- Chen H-J, Tsai Y-J, Chen W-S, Huang G-J, Huang S-S, Lin Y-H (2010) Ethephon-mediated effects on leaf senescence are affected by reduced glutathione and EGTA in sweet potato detached leaves. *Botanical Studies* 51, 171–181.
- Claussen W, Hawker JS, Loveys BR (1985) Sucrose synthase activity, invertase activity, net photosynthetic rates and carbohydrate content of detached leaves of eggplants as affected by attached stems and shoots (sinks). *Journal of Plant Physiology* 119, 123–131. [https://doi.org/10.1016/s0176-1617\(85\)80171-1](https://doi.org/10.1016/s0176-1617(85)80171-1).
- Dent RM, Cowan AK (1994) Absciscic acid-induced senescence in light-and dark-incubated detached leaves of barley. *South African Journal of Botany* 60, 340–344. [https://doi.org/10.1016/s0254-6299\(16\)30589-0](https://doi.org/10.1016/s0254-6299(16)30589-0).
- Dolui TK, Jana S (1988) Effects of phytohormones on some biochemical parameters during dark induced leaf senescence of *Sechium edule* on Darjeeling hill of the Eastern Himalayas. *Biologia Plantarum* 30, 379–383. <https://doi.org/10.1007/bf02878194>.
- Drew EA (1983) Physiology of *Laminaria*. I. Use of excised lamina discs in short and long term experiments. *Marine Ecology* 226, 259–270. <https://doi.org/10.1111/j.1439-0485.1983.tb00297.x>.
- Embry JL, Nothnagel EA (1988) Leaf development and senescence in *Panicum miliaceum* L., a cereal with a short seed-to-seed cycle. *Plant Science* 55, 129–136. [https://doi.org/10.1016/0168-9452\(88\)90168-9](https://doi.org/10.1016/0168-9452(88)90168-9).
- Embry JL, Nothnagel EA (1994) Photosynthetic light-harvesting during leaf senescence in *Panicum miliaceum*. *Plant Science* 95, 141–152. [https://doi.org/10.1016/0168-9452\(94\)90088-4](https://doi.org/10.1016/0168-9452(94)90088-4).
- Even-Chen Z, Atsmon D, Itai C (1978) Hormonal aspects of senescence in detached tobacco leaves. *Physiologia Plantarum* 44, 377–382. <https://doi.org/10.1111/j.1399-3054.1978.tb01641.x>.
- Frontier N, de Bettignies F, Foggo A, Davoult D (2021) Sustained productivity and respiration of degrading kelp detritus in the shallow benthos: detached or broken, but not dead. *Marine Environmental Research* 166, 105277. <https://doi.org/10.1016/j.marenvres.2021.105277>.
- Frontier N, Mulas M, Foggo A, Smale DA (2022) The influence of light and temperature on detritus degradation rates for kelp species with contrasting thermal affinities. *Marine Environmental Research* 173, 105529. <https://doi.org/10.1016/j.marenvres.2021.105529>.
- Gapper NE, Coupe SA, McKenzie MJ, Scott RW, Christey MC, Lill RE, McManus MT, Jameson PE (2005) Senescence-associated down-regulation of 1-aminocyclopropane-1-carboxylate (ACC) oxidase delays harvest-induced senescence in broccoli. *Functional Plant Biology* 32, 891–901. <https://doi.org/10.1071/fp05076>.

- Gauthier M-M, Jacobs DF (2018) Reductions in net photosynthesis and stomatal conductance vary with time since leaf detachment in three deciduous angiosperms. *Trees* 32, 1247–1252. <https://doi.org/10.1007/s00468-018-1706-z>.
- Gi E, Cho S-H, Kim S-H, Kang K, Paek N-C (2024) Rice ONAC016 promotes leaf senescence through abscisic acid signaling pathway involving OsNAP. *The Crop Journal* 12, 709–720. <https://doi.org/10.1016/j.cj.2024.02.009>.
- Grover A, Sabat SC, Mohanty P (1986) Effect of temperature on photosynthetic activities of senescing detached wheat leaves. *Plant and Cell Physiology* 27, 117–126. <https://doi.org/10.1093/oxfordjournals.pcp.a077071>.
- Grover A, Sabat SC, Mohanty P (1987) Does the loss of leaf chlorophyll during senescence of primary wheat leaf arise due to loss in chloroplast number or chlorophyll content. *Biochemie und Physiologie der Pflanzen* 182, 481–484. [https://doi.org/10.1016/s0015-3796\(87\)80078-1](https://doi.org/10.1016/s0015-3796(87)80078-1).
- Gut H, Rutz C, Matile P, Thomas H (1987) Leaf senescence in a non-yellowing mutant of *Festuca pratensis*: degradation of carotenoids. *Physiologia Plantarum* 70, 659–663. <https://doi.org/10.1111/j.1399-3054.1987.tb04321.x>.
- Hilditch P, Thomas H, Rogers L (1986) Leaf senescence in a non-yellowing mutant of *Festuca pratensis*: photosynthesis and photosynthetic electron transport. *Planta* 167, 146–151. <https://doi.org/10.1007/bf00446382>.
- Housley TL, Pollock CJ (1985) Photosynthesis and carbohydrate metabolism in detached leaves of *Lolium temulentum* L. *New Phytologist* 99, 499–507. <https://doi.org/10.1111/j.1469-8137.1985.tb03678.x>.
- Huang J, Han B, Xu S, Zhou M, Shen W (2011) Heme oxygenase-1 is involved in the cytokinin-induced alleviation of senescence in detached wheat leaves during dark incubation. *Journal of Plant Physiology* 168, 768–775. <https://doi.org/10.1016/j.jplph.2010.10.010>.
- Huang M, Feng L, He Y, Zhu Z (2006) Study on changing rules of chlorophyll concentration of detached canola leaves. *Fourth International Conference on Photonics and Imaging in Biology and Medicine* 6047, 604740. <https://doi.org/10.1117/12.710949>.
- Hukmani P, Tripathy BC (1994) Chlorophyll biosynthetic reactions during senescence of excised barley (*Hordeum vulgare* L. cv IB 65) leaves. *Plant Physiology* 105, 1295–1300. <https://doi.org/10.1104/pp.105.4.1295>.
- Jana S, Choudhuri MA (1979) Photosynthetic, photorespiratory and respiratory behaviour of three submersed aquatic angiosperms. *Aquatic Botany* 7, 13–19. [https://doi.org/10.1016/0304-3770\(79\)90003-2](https://doi.org/10.1016/0304-3770(79)90003-2).
- Jana S, Choudhuri MA (1980) Senescence in submerged aquatic angiosperms: changes in intact and isolated leaves during aging. *New Phytologist* 86, 191–198. <https://doi.org/10.1111/j.1469-8137.1980.tb03188.x>.
- Jana S, Choudhuri MA (1982a) Changes occurring during aging and senescence in a submerged aquatic angiosperm (*Potamogeton pectinatus*). *Physiologia Plantarum* 55, 356–360. <https://doi.org/10.1111/j.1399-3054.1982.tb00305.x>.
- Jana S, Choudhuri MA (1982b) Glycolate metabolism of three submersed aquatic angiosperms during ageing. *Aquatic Botany* 12, 345–354. [https://doi.org/10.1016/0304-3770\(82\)90026-2](https://doi.org/10.1016/0304-3770(82)90026-2).
- Jana S, Choudhuri MA (1982c) Changes in the activities of ribulose 1, 5-bisphosphate and phosphoenolpyruvate carboxylases in submersed aquatic angiosperms during aging. *Plant Physiology* 70, 1125–1127. <https://doi.org/10.1104/pp.70.4.1125>.
- Jana S, Choudhuri MA (1984a) Synergistic effects of heavy metal pollutants on senescence in submerged aquatic plants. *Water, Air, & Soil Pollution* 21, 351–357. <https://doi.org/10.1007/bf00163635>.
- Jana S, Choudhuri MA (1984b) Effects of plant growth regulators on Hill activity of submerged aquatic plants during induced senescence. *Aquatic Botany* 18, 371–376. [https://doi.org/10.1016/0304-3770\(84\)90057-3](https://doi.org/10.1016/0304-3770(84)90057-3).

- Jana S (1988) Glycolate metabolism of wheat and rice leaves during senescence and under the influence of growth regulators. *Biologia Plantarum* 30, 30–33.
<https://doi.org/10.1007/bf02876420>.
- Jia HS, Han YQ, Li DQ (2003) Photoinhibition and active oxygen species production in detached apple leaves during dehydration. *Photosynthetica* 41, 151–156.
<https://doi.org/10.1023/a:1025889219115>.
- Jiang W, Sheng Q, Zhou X-J, Zhang M-J, Liu X-J (2002) Regulation of detached coriander leaf senescence by 1-methylcyclopropene and ethylene. *Postharvest Biology and Technology* 26, 339–345. [https://doi.org/10.1016/s0925-5214\(02\)00068-6](https://doi.org/10.1016/s0925-5214(02)00068-6).
- Kang K, Kim YS, Park S, Back K (2009) Evaluation of light-harvesting complex proteins as senescence-related protein markers in detached rice leaves. *Photosynthetica* 47, 638–640.
<https://doi.org/10.1007/s11099-009-0093-5>.
- Kao CH, Yang SF (1983) Role of ethylene in the senescence of detached rice leaves. *Plant Physiology* 73, 881–885. <https://doi.org/10.1104/pp.73.4.881>.
- Kar M, Feierabend J (1984) Metabolism of activated oxygen in detached wheat and rye leaves and its relevance to the initiation of senescence. *Planta* 160, 385–391.
<https://doi.org/10.1007/bf00429753>.
- Kar M, Streb P, Hertwig B, Feierabend J (1993) Sensitivity to photodamage increases during senescence in excised leaves. *Journal of Plant Physiology* 141, 538–544.
[https://doi.org/10.1016/s0176-1617\(11\)80453-0](https://doi.org/10.1016/s0176-1617(11)80453-0).
- Kar RK, Choudhuri MA (1985) Senescence of *Hydrilla* leaves in light and darkness. *Physiologia Plantarum* 63, 225–230. <https://doi.org/10.1111/j.1399-3054.1985.tb01907.x>.
- Kar S, Montague DT, Villanueva-Morales A (2021a) Measurement of photosynthesis in excised leaves of ornamental trees: a novel method to estimate leaf level drought tolerance and increase experimental sample size. *Trees* 35, 889–905. <https://doi.org/10.1007/s00468-021-02088-w>.
- Kar S, Montague T, Villanueva-Morales A, Hellman E (2021b) Measurement of gas exchange on excised grapevine leaves does not differ from *in situ* leaves, and potentially shortens sampling time. *Applied Sciences* 11, 3644. <https://doi.org/10.3390/app11083644>.
- Karatas I, Ozturk L, Ersahin Y, Okatan Y (2010) Effects of auxin on photosynthetic pigments and some enzyme activities during dark-induced senescence of *Tropaeolum* leaves. *Pakistan Journal of Botany* 42, 1881–1888.
- Kato MC, Hikosaka K, Hirose T (2002) Leaf discs floated on water are different from intact leaves in photosynthesis and photoinhibition. *Photosynthesis Research* 72, 65–70.
<https://doi.org/10.1023/a:1016097312036>.
- Knauer GA, Ayers AV (1977) Changes in carbon, nitrogen, adenosine triphosphate, and chlorophyll *a* in decomposing *Thalassia testudinum* leaves. *Limnology and Oceanography* 22, 408–414. <https://doi.org/10.4319/lo.1977.22.3.0408>.
- Kreslavski VD, Lankin AV, Vasilyeva GK, Luybimov VY, Semenova GN, Schmitt FJ, Friedrich T, Allakhverdiev SI (2014) Effects of polyaromatic hydrocarbons on photosystem II activity in pea leaves. *Plant Physiology and Biochemistry* 81, 135–142.
<https://doi.org/10.1016/j.plaphy.2014.02.020>.
- Landolt R, Matile P (1990) Glyoxisome-like microbodies in senescent spinach leaves. *Plant Science* 72, 159–163. [https://doi.org/10.1016/0168-9452\(90\)90078-3](https://doi.org/10.1016/0168-9452(90)90078-3).
- Lewington RJ, Talbot M, Simon EW (1967) The yellowing of attached and detached cucumber cotyledons. *Journal of Experimental Botany* 18, 526–534.
<https://doi.org/10.1093/jxb/18.3.526>.
- Lewington RJ, Simon EW (1969) The effect of light on the senescence of detached cucumber cotyledons. *Journal of Experimental Botany* 20, 138–144.
<https://doi.org/10.1093/jxb/20.1.138>.
- Liu L, Li H, Zeng H, Cai Q, Zhou X, Yin C (2016) Exogenous jasmonic acid and cytokinin antagonistically regulate rice flag leaf senescence by mediating chlorophyll degradation,

- membrane deterioration, and senescence-associated genes expression. *Journal of Plant Growth Regulation* 35, 366–376. <https://doi.org/10.1007/s00344-015-9539-0>.
- Lovell PH, Illsley ANN, Moore KG (1972) The effects of light intensity and sucrose on root formation, photosynthetic ability, and senescence in detached cotyledons of *Sinapis alba* L. and *Raphanus sativus* L. *Annals of Botany* 36, 123–134. <https://doi.org/10.1093/oxfordjournals.aob.a084565>.
- Martin C, Thimann KV (1972) The role of protein synthesis in the senescence of leaves: I. The formation of protease. *Plant Physiology* 49, 64–71. <https://doi.org/10.1104/pp.49.1.64>.
- Miller BL, Huffaker RC (1985) Differential induction of endoproteinases during senescence of attached and detached barley leaves. *Plant Physiology* 78, 442–446. <https://doi.org/10.1104/pp.78.3.442>.
- Mishra D, Kar M (1973) Effect of benzimidazole and nickel ions in the control of *Oryza sativa* leaf senescence by red and far-red light. *Phytochemistry* 12, 1521–1522. [https://doi.org/10.1016/0031-9422\(73\)80360-7](https://doi.org/10.1016/0031-9422(73)80360-7).
- Misr AN, Meena M (1986) Effect of temperature on senescing detached rice leaves: I. Photoelectron transport activity of chloroplasts. *Plant Science* 46, 1–4. [https://doi.org/10.1016/0168-9452\(86\)90123-8](https://doi.org/10.1016/0168-9452(86)90123-8).
- Misra AN, Biswal UC (1980) Effect of phytohormones on chlorophyll degradation during aging of chloroplasts in vivo and in vitro. *Protoplasma* 105, 1–8. <https://doi.org/10.1007/bf01279845>.
- Misra AN, Biswal UC (1982) Differential changes in the electron transport properties of thylakoid membranes during aging of attached and detached leaves, and of isolated chloroplasts. *Plant, Cell & Environment* 5, 27–30. <https://doi.org/10.1111/1365-3040.ep11587573>.
- Momose T, Ozeki Y (2013) Regulatory effect of stems on sucrose-induced chlorophyll degradation and anthocyanin synthesis in *Egeria densa* leaves. *Journal of Plant Research* 126, 859–867. <https://doi.org/10.1007/s10265-013-0581-3>.
- Mondal R, Choudhuri MA (1981) Role of hydrogen peroxide in senescence of excised leaves of rice and maize. *Biochemie und Physiologie der Pflanzen* 176, 700–709. [https://doi.org/10.1016/s0015-3796\(81\)80055-8](https://doi.org/10.1016/s0015-3796(81)80055-8).
- Montague T, Villanueva-Morales A, Khan MAR, Wallace R, Panta S (2024) Comparing the accuracy and efficiency of leaf gas exchange measurements of excised and nonexcised strawberry (*Fragaria × ananassa* Duch. 'Camino Real') leaves. *HortTechnology* 34, 728–737. <https://doi.org/10.21273/horttech05516-24>.
- Nedunchezian N, Ravindran KC, Kulandaivelu G (1995) Changes in photosynthetic apparatus during dark incubation of detached leaves from control and ultraviolet-B treated *Vigna* seedlings. *Biologia Plantarum* 37, 341–348. <https://doi.org/10.1007/bf02913238>.
- Nock LP, Rogers LJ, Thomas H (1992) Metabolism of protein and chlorophyll in leaf tissue of *Festuca pratensis* during chloroplast assembly and senescence. *Phytochemistry* 31, 1465–1470. [https://doi.org/10.1016/0031-9422\(92\)83088-g](https://doi.org/10.1016/0031-9422(92)83088-g).
- Ogwenio J-O, Song X-S, Hu W-H, Shi K, Zhou Y-H, Yu J-Q (2009) Detached leaves of tomato differ in their photosynthetic physiological response to moderate high and low temperature stress. *Scientia Horticulturae* 123, 17–22. <https://doi.org/10.1016/j.scienta.2009.07.011>.
- Oh M-H, Moon Y-H, Lee C-H (2003) Increased stability of LHCII by aggregate formation during dark-induced leaf senescence in the *Arabidopsis* mutant, ore10. *Plant and Cell Physiology* 44, 1368–1377. <https://doi.org/10.1093/pcp/pcg170>.
- Okada K, Inoue Y, Satoh K, Katoh S (1992) Effects of light on degradation of chlorophyll and proteins during senescence of detached rice leaves. *Plant and Cell Physiology* 33, 1183–1191. <https://doi.org/10.1093/oxfordjournals.pcp.a078372>.
- Pellikaan GC (1982) Decomposition processes of eelgrass, *Zostera marina* L. *Hydrobiological Bulletin* 16, 83–92. <https://doi.org/10.1007/bf02255416>.

- Philosoph-Hadas S, Meir S, Aharoni N (1991) Effect of wounding on ethylene biosynthesis and senescence of detached spinach leaves. *Physiologia Plantarum* 83, 241–246. <https://doi.org/10.1111/j.1399-3054.1991.tb02148.x>.
- Philosoph-Hadas S, Meir S, Akiri B, Kanner J (1994a) Oxidative defense systems in leaves of three edible herb species in relation to their senescence rates. *Journal of Agricultural and Food Chemistry* 42, 2376–2381. <https://doi.org/10.1021/jf00047a003>.
- Philosoph-Hadas S, Meir S, Aharoni N (1994b) Role of ethylene in senescence of watercress leaves. *Physiologia Plantarum* 90, 553–559. <https://doi.org/10.1111/j.1399-3054.1994.tb08814.x>.
- Reyes-Arribas T, Barrett JE, Huber DJ, Nell TA, Clark DG (2001) Leaf senescence in a non-yellowing cultivar of chrysanthemum (*Dendranthema grandiflora*). *Physiologia Plantarum* 111, 540–544. <https://doi.org/10.1034/j.1399-3054.2001.1110415.x>.
- Richardson SD (1957) The effect of leaf age on the rate of photosynthesis in detached leaves of tree seedlings. *Acta Botanica Neerlandica* 6, 445–457. <https://doi.org/10.1111/j.1438-8677.1957.tb00592.x>.
- Rodoni S, Schellenberg M, Matile P (1998) Chlorophyll breakdown in senescing barley leaves as correlated with phaeophorbide a oxygenase activity. *Journal of Plant Physiology* 152, 139–144. [https://doi.org/10.1016/s0176-1617\(98\)80124-7](https://doi.org/10.1016/s0176-1617(98)80124-7).
- Rodríguez MT, González MP, Linares JM (1987) Degradation of chlorophyll and chlorophyllase activity in senescing barley leaves. *Journal of Plant Physiology* 129, 369–374. [https://doi.org/10.1016/s0176-1617\(87\)80094-9](https://doi.org/10.1016/s0176-1617(87)80094-9).
- Rothäusler E, Gómez I, Hinojosa IA, Karsten U, Miranda L, Tala F, Thiel M (2011) Kelp rafts in the Humboldt Current: Interplay of abiotic and biotic factors limit their floating persistence and dispersal potential. *Limnology and Oceanography* 56, 1751–1763. <https://doi.org/10.4319/lo.2011.56.5.1751>.
- Sabater B, Rodríguez MT (1978) Control of chlorophyll degradation in detached leaves of barley and oat through effect of kinetin on chlorophyllase levels. *Physiologia Plantarum* 43, 274–276. <https://doi.org/10.1111/j.1399-3054.1978.tb02577.x>.
- Sağlam-Çağ S (2007) The effect of epibrassinolide on senescence in wheat leaves. *Biotechnology & Biotechnological Equipment* 21, 63–65. <https://doi.org/10.1080/13102818.2007.10817415>.
- Shafique S, Siddiqui PJS, Aziz RA, Burhan Z, Mansoor SN (2010) Weight loss and changes in organic, inorganic and chlorophyll contents in three species of seaweeds during decomposition. *Pakistan Journal of Botany* 42, 2599–2604.
- Sharma R, Biswal UC (1976) Effect of kinetin and light on peroxidase activity of detached barley leaves. *Zeitschrift für Pflanzenphysiologie* 78, 169–172. [https://doi.org/10.1016/s0044-328x\(78\)80188-3](https://doi.org/10.1016/s0044-328x(78)80188-3).
- Shaw M, Bhattacharya PK, Quick WA (1965) Chlorophyll, protein, and nucleic acid levels in detached, senescing wheat leaves. *Canadian Journal of Botany* 43, 739–746. <https://doi.org/10.1139/b65-083>.
- Singh N, Mishra D (1975) The effect of benzimidazole and red light on the senescence of detached leaves of *Oryza sativa* cv. Ratna. *Physiologia Plantarum* 34, 67–74. <https://doi.org/10.1111/j.1399-3054.1975.tb01858.x>.
- Špundová M, Popelková H, Ilík P, Skotnica J, Novotný R, Nauš J (2003) Ultra-structural and functional changes in the chloroplasts of detached barley leaves senescing under dark and light conditions. *Journal of Plant Physiology* 160, 1051–1058. <https://doi.org/10.1078/0176-1617-00902>.
- Špundová M, Strzałka K, Nauš J (2005) Xanthophyll cycle activity in detached barley leaves senescing under dark and light. *Photosynthetica* 43, 117–124. <https://doi.org/10.1007/s11099-005-7124-7>.
- Srivastava BI, Ware G (1965) The effect of kinetin on nucleic acids and nucleases of excised barley leaves. *Plant Physiology* 40, 62. <https://doi.org/10.1104/pp.40.1.62>.

- Strother S, Vatta R (1986) The effects of light, cycloheximide and incubation medium on the senescence of detached seagrass leaves. *Biologia Plantarum* 28, 211–218. <https://doi.org/10.1007/bf02894599>.
- Subhan D, Murthy SDS (2000) Synergistic effect of AlCl₃ and kinetin on chlorophyll and protein contents and photochemical activities in detached wheat primary leaves during dark incubation. *Photosynthetica* 38, 211–214. <https://doi.org/10.1023/a:1007261614200>.
- Subhan D, Murthy SDS (2001a) Senescence retarding effect of metal ions: pigment and protein contents and photochemical activities of detached primary leaves of wheat. *Photosynthetica* 39, 53–58. <https://doi.org/10.1023/a:1012487718114>.
- Subhan D, Murthy SDS (2001b) Effect of polyamines on chlorophyll and protein contents, photochemical activity, and energy transfer in detached wheat leaves during dark incubation. *Biologia Plantarum* 44, 529–533. <https://doi.org/10.1023/a:1013734502112>.
- Tala F, Velásquez M, Mansilla A, Macaya EC, Thiel M (2016) Latitudinal and seasonal effects on short-term acclimation of floating kelp species from the South-East Pacific. *Journal of Experimental Marine Biology and Ecology* 483, 31–41. <https://doi.org/10.1016/j.jembe.2016.06.003>.
- Tala F, Penna-Díaz MA, Luna-Jorquera G, Rothäusler E, Thiel M (2017) Daily and seasonal changes of photobiological responses in floating bull kelp *Durvillaea antarctica* (Chamisso) Hariot (Fucales: Phaeophyceae). *Phycologia* 56, 271–283. <https://doi.org/10.2216/16-93.1>.
- Tetley RM, Thimann KV (1974) The metabolism of oat leaves during senescence: I. Respiration, carbohydrate metabolism, and the action of cytokinins. *Plant Physiology* 54, 294–303. <https://doi.org/10.1104/pp.54.3.294>.
- Thimann KV, Tetley RR, Van Thanh T (1974) The metabolism of oat leaves during senescence: II. Senescence in leaves attached to the plant. *Plant Physiology* 54, 859–862. <https://doi.org/10.1104/pp.54.6.859>.
- Thimann KV, Tetley RM, Krivak BM (1977) Metabolism of oat leaves during senescence: V. Senescence in light. *Plant Physiology* 59, 448–454. <https://doi.org/10.1104/pp.59.3.448>.
- Thimann KV, Satler SO (1979) Relation between leaf senescence and stomatal closure: senescence in light. *Proceedings of the National Academy of Sciences* 76, 2295–2298. <https://doi.org/10.1073/pnas.76.5.2295>.
- Thimann KV (1985) The senescence of detached leaves of *Tropaeolum*. *Plant Physiology* 79, 1107–1110. <https://doi.org/10.1104/pp.79.4.1107>.
- Thomas H, Stoddart JL (1975) Separation of chlorophyll degradation from other senescence processes in leaves of a mutant genotype of meadow fescue (*Festuca pratensis* L.). *Plant Physiology* 56, 438–441. <https://doi.org/10.1104/pp.56.3.438>.
- Thomas H (1987) Sid: a Mendelian locus controlling thylakoid membrane disassembly in senescing leaves of *Festuca pratensis*. *Theoretical and Applied Genetics* 73, 551–555. <https://doi.org/10.1007/bf00289193>.
- Turner WB, Bidwell RGS (1965) Rates of photosynthesis in attached and detached bean leaves, and the effect of spraying with indoleacetic acid solution. *Plant Physiology* 40, 446. <https://doi.org/10.1104/pp.40.3.446>.
- van Hees DH, Olsen YS, Mattio L, Ruiz-Montoya L, Wernberg T, Kendrick GA (2019) Cast adrift: physiology and dispersal of benthic *Sargassum spinuligerum* in surface rafts. *Limnology and Oceanography* 64, 526–540. [10.1002/lno.11057](https://doi.org/10.1002/lno.11057).
- Veierskov B, Kirk HG (1986) Senescence in oat leaf segments under hypobaric conditions. *Physiologia Plantarum* 66, 283–287. <https://doi.org/10.1111/j.1399-3054.1986.tb02421.x>.
- Veierskov B (1987) Irradiance-dependent senescence of isolated leaves. *Physiologia Plantarum* 71, 316–320. <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1399-3054.1987.tb04349.x>.

- Vlčková A, Špundová M, Kotabová E, Novotný R, Doležal K, Nauš J (2006) Protective cytokinin action switches to damaging during senescence of detached wheat leaves in continuous light. *Physiologia Plantarum* 126, 257–267. <https://doi.org/10.1111/j.1399-3054.2006.00593.x>.
- Voronin PY, Fedoseeva GP (2012) Stomatal control of photosynthesis in detached leaves of woody and herbaceous plants. *Russian Journal of Plant Physiology* 59, 281–286. <https://doi.org/10.1134/s1021443712020173>.
- Wang P, Yin L, Liang D, Li C, Ma F, Yue Z (2012) Delayed senescence of apple leaves by exogenous melatonin treatment: toward regulating the ascorbate-glutathione cycle. *Journal of Pineal Research* 53, 11–20. <https://doi.org/10.1111/j.1600-079x.2011.00966.x>.
- Wen B, Li C, Fu X, Li D, Li L, Chen X, Wu H, Cui X, Zhang X, Shen H et al. (2019) Effects of nitrate deficiency on nitrate assimilation and chlorophyll synthesis of detached apple leaves. *Plant Physiology and Biochemistry* 142, 363–371. <https://doi.org/10.1016/j.plaphy.2019.07.007>.
- Wright LS, Foggo A (2021) Photosynthetic pigments of co-occurring Northeast Atlantic *Laminaria* spp. are unaffected by decomposition. *Marine Ecology Progress Series* 678, 227–232. <https://doi.org/10.3354/meps13886>.
- Wright LS, Pessarrodona A, Foggo A (2022) Climate-driven shifts in kelp forest composition reduce carbon sequestration potential. *Global Change Biology* 28, 5514–5531. <https://doi.org/10.1111/gcb.16299>.
- Wright LS, Kregting L (2023) Genus-specific response of kelp photosynthetic pigments to decomposition. *Marine Biology* 170, 144. <https://doi.org/10.1007/s00227-023-04289-y>.
- Wright LS, Simpkins T, Filbee-Dexter K, Wernberg T (2024) Temperature sensitivity of detrital photosynthesis. *Annals of Botany* 133, 17–28. <https://doi.org/10.1093/aob/mcad167>.
- Xiao X, de Bettignies T, Olsen YS, Agusti S, Duarte CM, Wernberg T (2015) Sensitivity and acclimation of three canopy-forming seaweeds to UVB radiation and warming. *PLoS One* 10, e0143031. <https://doi.org/10.1371/journal.pone.0143031>.
- Yang J, Worley E, Udvardi M (2014) A NAP-AAO3 regulatory module promotes chlorophyll degradation via ABA biosynthesis in *Arabidopsis* leaves. *The Plant Cell* 26, 4862–4874. <https://doi.org/10.1105/tpc.114.133769>.
- Yoshida Y (1970) Control of chloroplast senescence of detached *Elodea* leaves. *The Botanical Magazine (Shokubutsugaku Zasshi)* 83, 137–143. <https://doi.org/10.15281/jplantres1887.83.137>.
- Zhang Z, Li G, Gao H, Zhang L, Yang C, Liu P, Meng Q (2012) Characterization of photosynthetic performance during senescence in stay-green and quick-leaf-senescence *Zea mays* L. inbred lines. *PLoS One* 7, e42936. <https://doi.org/10.1371/journal.pone.0042936>.
- Zhang Z, Guo J, Yang C, Ma Y, Gao S, Liu Q, Zhang J, Ma F, Li C (2023) Exogenous dopamine delays the senescence of detached *Malus hupehensis* leaves by reducing phytohormone signalling and sugar degradation. *Scientia Horticulturae* 319, 112151. <https://doi.org/10.1016/j.scienta.2023.112151>.
- Zhao L, Zhang H, Zhang B, Bai X, Zhou C (2012) Physiological and molecular changes of detached wheat leaves in responding to various treatments. *Journal of Integrative Plant Biology* 54, 567–576. <https://doi.org/10.1111/j.1744-7909.2012.01139.x>.